

Environmental Externality: Definition & Internalisation

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External Cost

(Definition, Optimal level, Problem Solution)

- What Is A externality: Definition of Good Ones and Bad Ones?
 - Is There an Optimal Level of the externality ?
 - How to Achieve the Optimum ?
 - How to Internalise the externality ?

Literature:

- Verhoef, E. (1994), External Effects And Social Costs of Road Transport. Transportation Research Part A Vol. 28A, 4, 273-287
- Verhoef, E.T. (2002), Externalities, in: van der Bergh, J.C.J.M. (ed.), Handbook of Environmental and Resource Economics. Edward Elgar, Cheltenham, 197-214.

Textbook Example

upstream industry **discharges waste** to a river

→ loss of dissolved in **oxygen**

→ loss of **fish stock** in the river

→ financial and recreational (physical) **losses** downstream

→ **welfare loss** to anglers

if these losses are **not compensated**, the upstream industry will continue its activities as if the damage done downstream was irrelevant to them → **overproduction** in industry leads to **suboptimal allocation** of resources

Terminology

- terminology
 - the external effects
 - external +/- savings
 - technological external effects, in line with standard terminology of Viner (1931) or Scitovsky (1954) that concerns „**technological**“ externalities (as opposed to „pecuniary externalities“)
- the externalities are potentially „**Pareto relevant**“ (Buchanan and Stubblebine 1962)
 - if the costs of correcting for the market failure do **not exceed** the welfare gains to be obtained

Definitions

- An external costs exists when two conditions prevail>
 - an activity by **one** agent **causes a loss of welfare** to **another** agent
 - the loss of welfare **is uncompensated**

(Pearce and Turner 1991)

- „Externality is a form of **uncompensated effect** on a **utility function** or a **production set**.“

(Verhoef 1994)

- „...externality arises when the social or economic activities of one group of persons have an **impact on another group** and when that impacts is **not fully accounted**, or compensated for; by the first group“

(European Commission 2003)

Definitions cont'd.

- An externality is present whenever some individual's utility or production relationships include **real variables**, whose values are **chosen by others without particular attention to effects** on this individual's welfare... (Baumol and Oates 1988: 17-18)
- An external effect exists when an actor's (the receptor's) utility (or profit) function contains **a real variable** whose actual value depends on the behaviour of another actor (the supplier), who **does not take** these effects of his behaviour **into account** in his decision making process. (Verhoef 1994; 2002)
- The essential feature of an external effect [is] that the effect produced is **not a deliberate creation** but **an intended or incidental by-product** of some otherwise legitimate activity (Mishan, 1971; cit in: Verhoef 2002)
- ...the decision-maker whose activity affects others' utility levels or enters their production function, does not **receive (pay) compensation** for this activity an amount equal in value to the resulting benefits (or costs) to others... (Baumol and Oates 1988: 17-18)

Effect that does not qualify to be called as the Externalities

1] Is utility / profit affected ?

- the effect does not have any impact on utility of the production function
- (physical) impact on the environment

Effects that do not qualify as the Externalities

1] Is utility / profit affected ?

2] Was the impact (affecting another agent) **taken into account in decision-making** of its producer?

- the effect should not be taken into account in supplier's decisions
- distinguish *unpriced external relations* from the externality
 - *barter trade* [there is no external benefit]
 - *criminal (violent) activities* [receptor's utility level as determining variable]
 - *altruism, charity, goodwill* [receptor's pleasure enters into the supplier utility]
- individual welfare maximizing behavior is perfectly in line with Pareto optimality → policy action not required on the basis of efficiency
 - only in the case of social (equity) concerns

Effects that do not qualify as the Externalities

1] Is utility / profit affected ?

2] Was the impact taken into account in decision process?

3] Does the effect present real variable effect of rather **ordinary economic dependencies**?

- true (technological) externalities aim at **real variables**, i.e. excluding monetary transactions (see Viner, 1931; or Scitovski, 1954)
- distinguish the true externality from *pecuniary external effect*
 - presents ordinary economic dependencies acting through market
 - mutual **interrelations between sectors and agents**
 - does not lead to shifts in production or utility functions but merely to **movements along these functions**
 - **no intervention** is required to secure Pareto efficiency
 - pecuniary effects can not compensate the true external costs

Effects that do not qualify as the Externalities

1] Is utility / profit affected ?

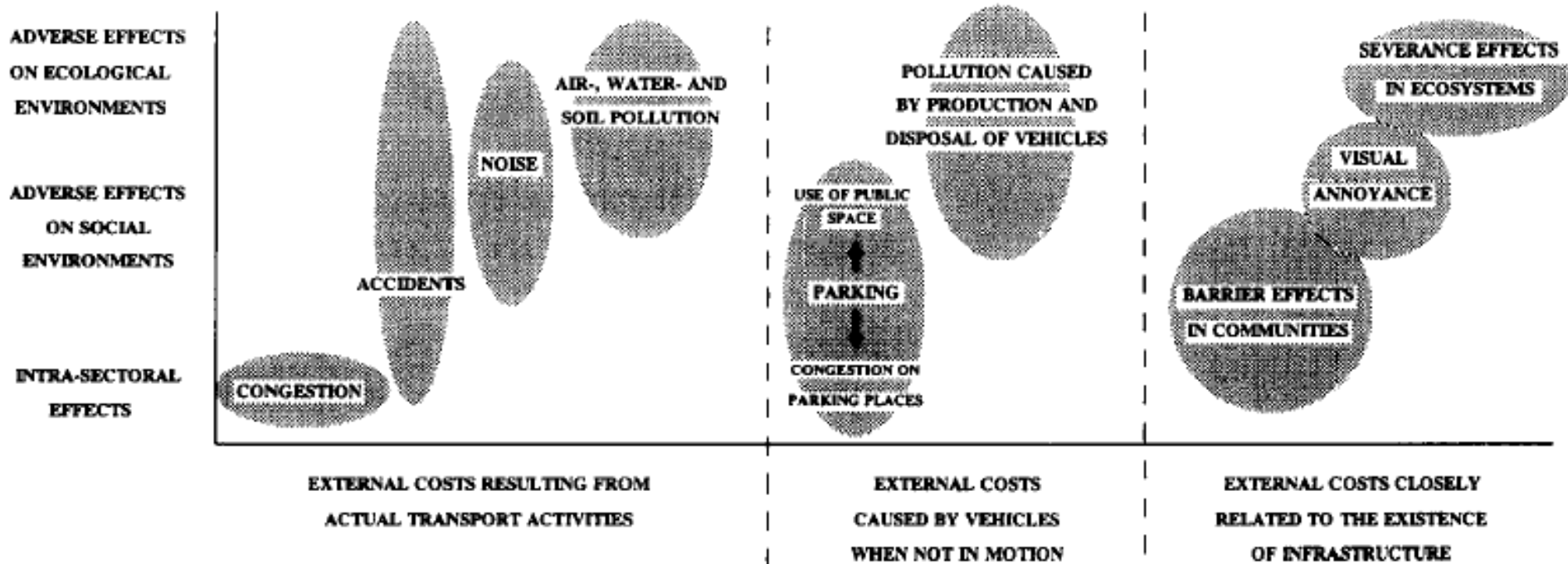
2] Was the impact taken into account in decision process?

3] Does the effect present pecuniary, rather than real variable?

4] Is it a case of other benefit involved?

- Induced benefits of some intervention such as **value added** or **employment** in sectors (e.g. vehicle production, repair services and fuel production involved by boosting transport infrastructure)
 - These effects do not provide economic rationale for moving from the optimal level
- Benefits provided by infrastructure extension such as **boosting econ dvlp.** and **regional employment**
- But econ effects due to infrastructure construction can be slightly positive if the infrastructure was missing, or negative in the case of redistribution among regions (Rietveld 1989)
 - these effects might be considered in CBA of road construction
 - investment might lead to max welfare only if transport has not been exceeding its optimal level

Externality due to transport



Verhoef 1994: 276

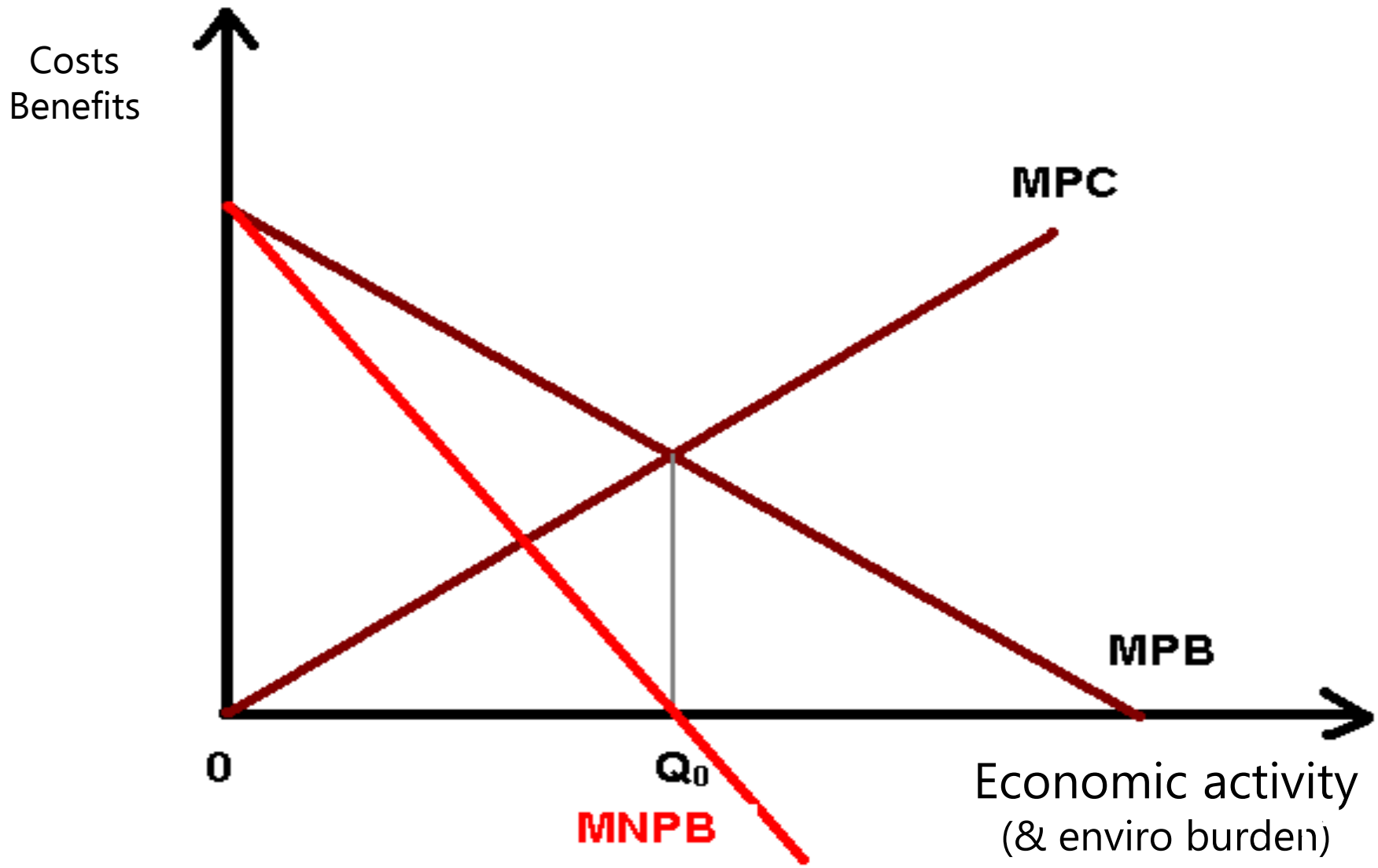
- First order effects
- Second order effects (e.g., congestions may lead to more pollution)

Externality classification

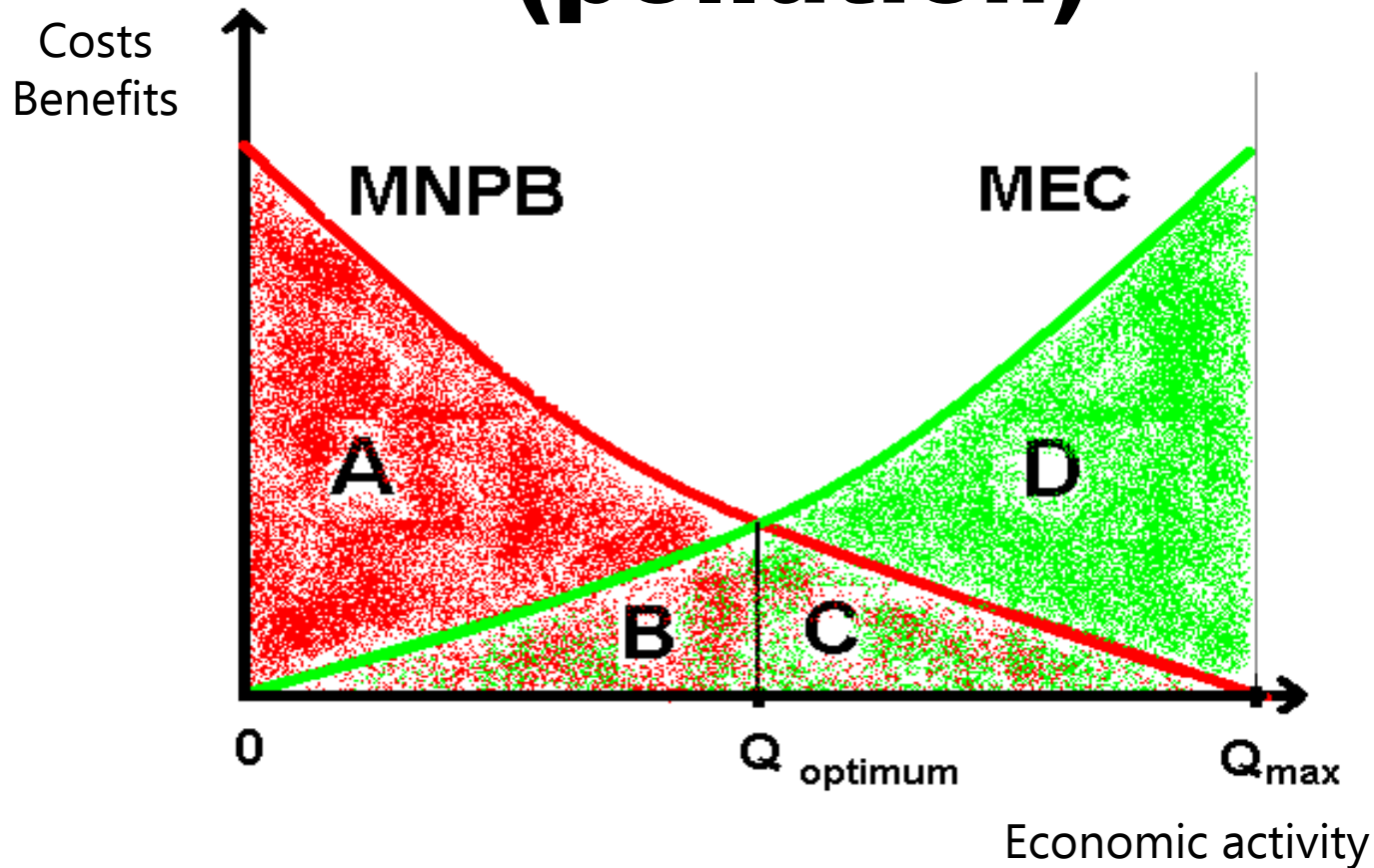
- ❑ **positive** versus **negative**
- ❑ **production** versus **consumption**
- ❑ **neighbourhood effects** (or effects of overflowing; with spatial dimension)
- ❑ **depletable** versus **undepletable** (also qualifies as a public good)
- ❑ **congestion externality** (agent is simultaneously a supplier as well as a recipient)
- ❑ **partial** versus **global** (according to its potential spatial impacts)
- ❑ **intergeneration** (related to equity, e.g. a resource depletion (viz Stiglitz))
- ❑ **x-dimensional** (e.g. fishermen; fisherman x boating; fisherman x boating x swimming...)
- ❑ **marginal** versus **inframarginal**

Is There An Optimal Level Of the
Externality?

Optimal level of externality

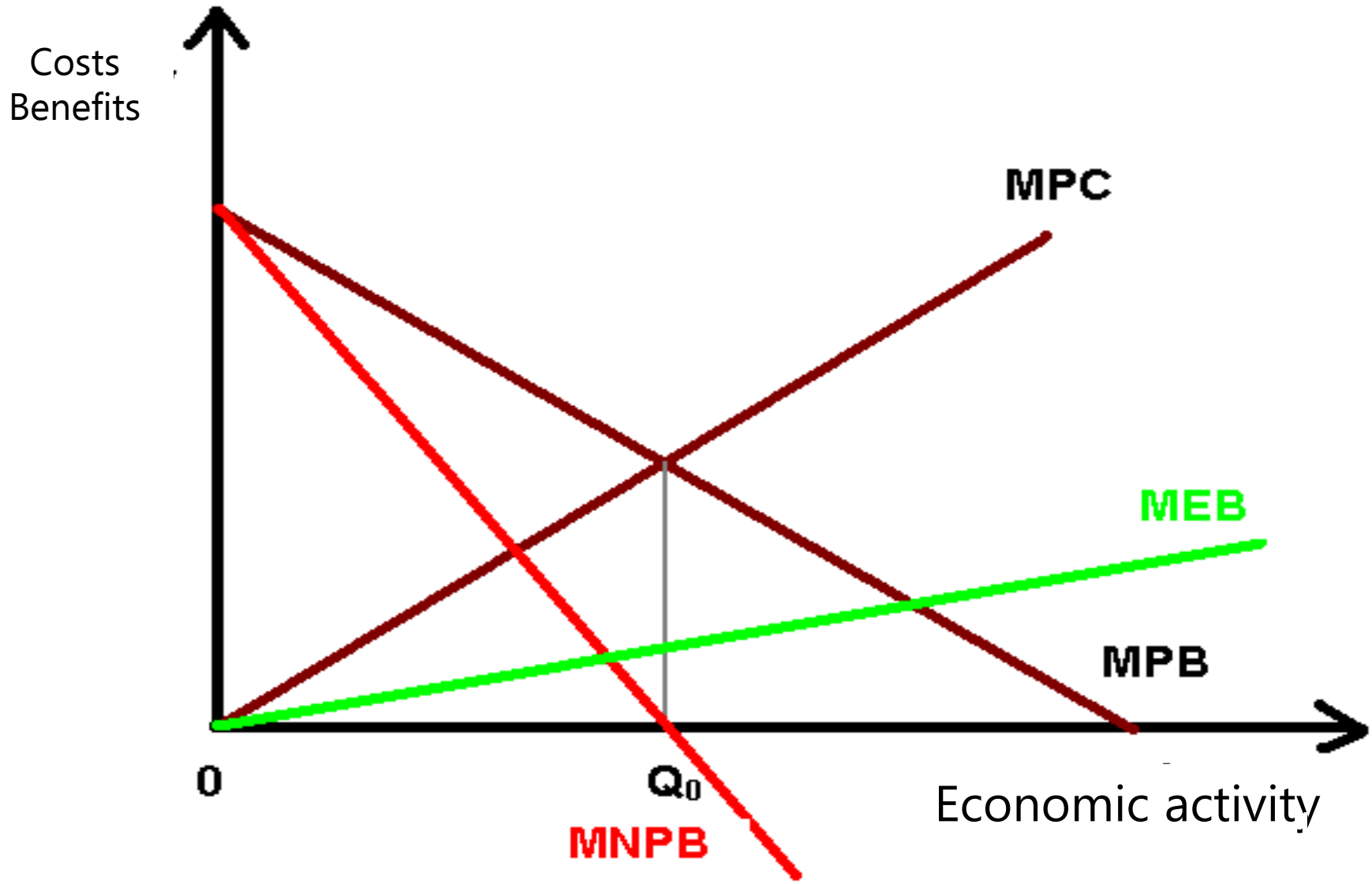


Optimal level of regulation (pollution)

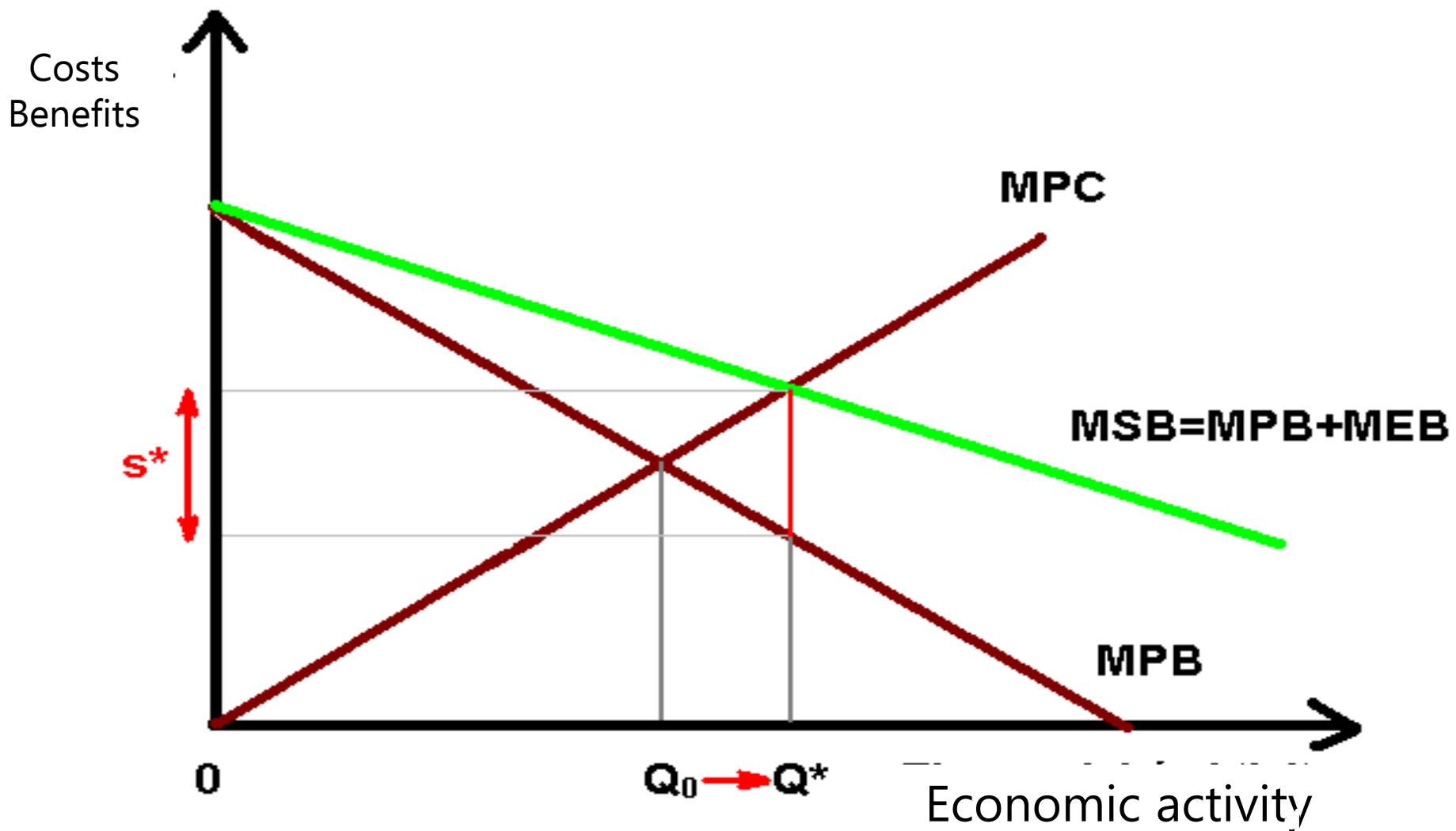


			<i>profit</i>	<i>damage</i>	
<i>no emission</i>	→	Social Welfare (0)	=	0	– 0 = 0
<i>no regulation</i>	→	Social Welfare (Q_{max})	=	(A+B+C)	– (B+C+D) = <u>A</u> – <u>D</u>
<i>optimum</i>	→	Social Welfare (Q_{opt})	=	(A+B)	– B = A

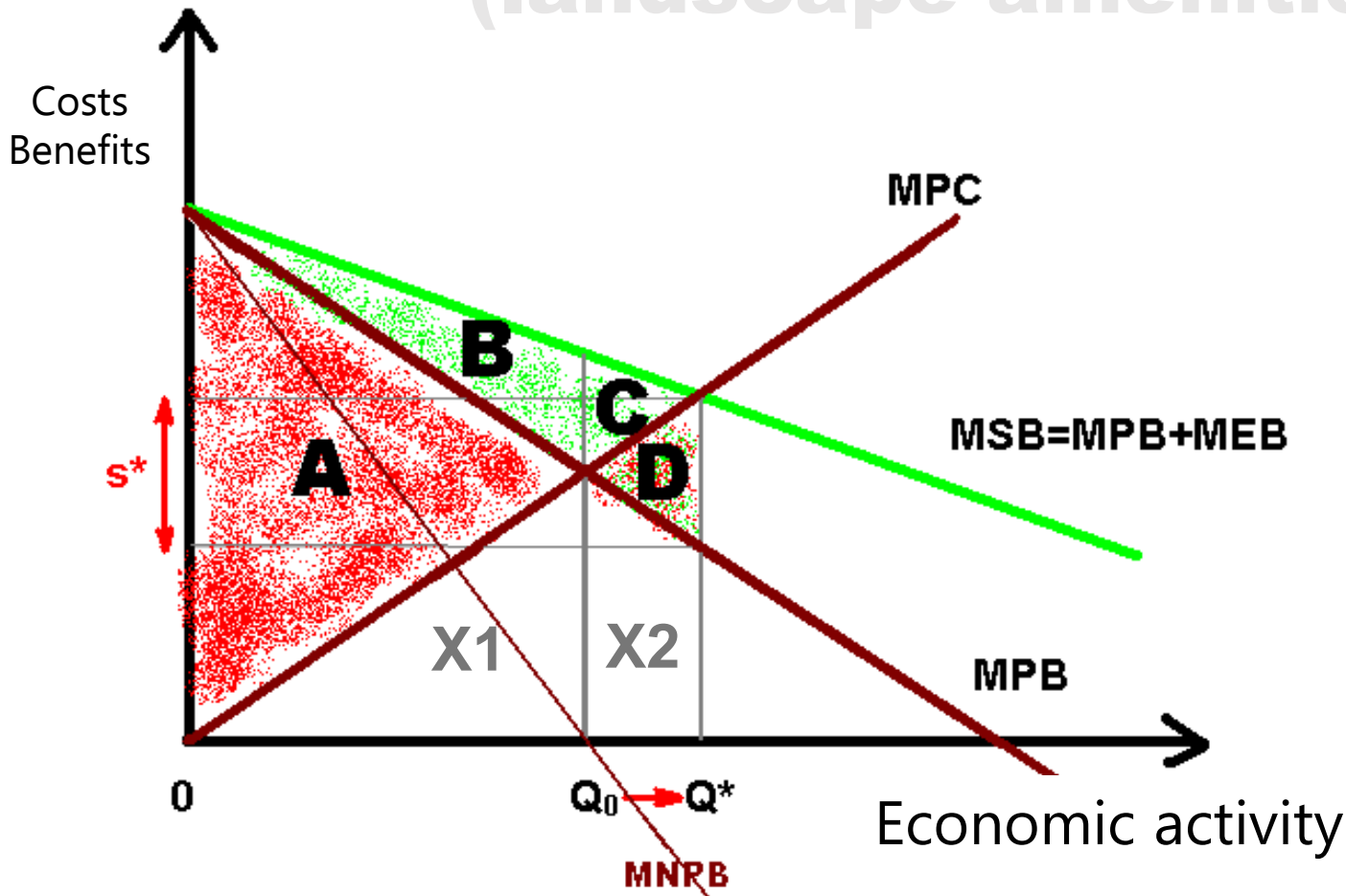
Optimal level of positive externality



Optimal level of positive externality



Optimal level of regulation (landscape amenities)



no output

no regulation

more than Q^*

less than Q^*

optimum

→ Social Welfare (0) = 0

→ Social Welfare (Q_0) = A + B

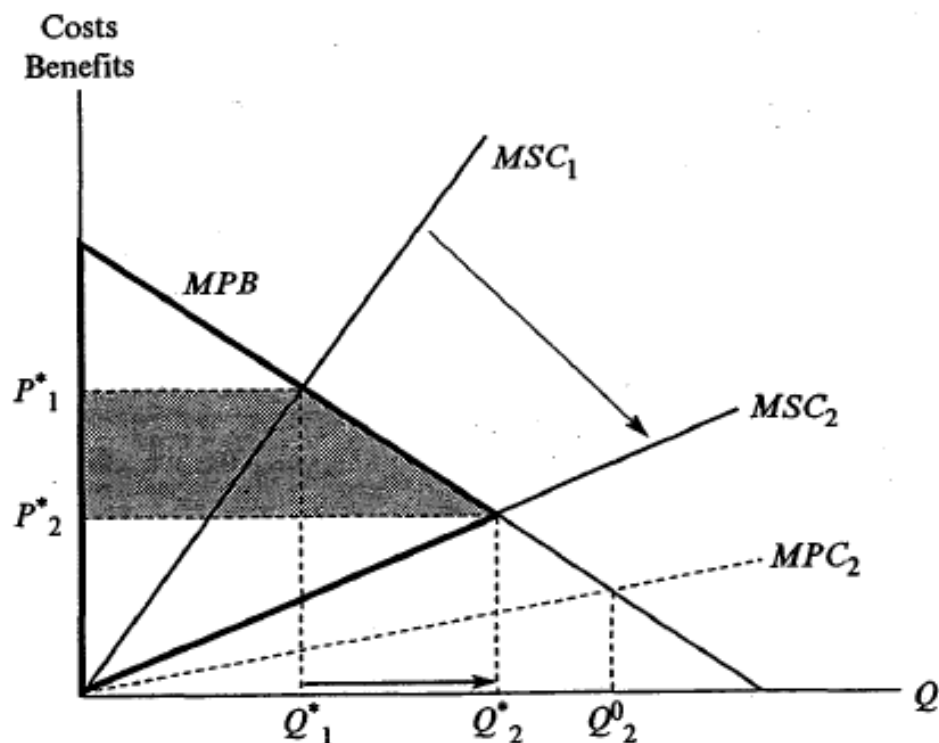
→ Social Welfare (Q_+) = (A + B) + C + (MSB[q_+] - MPC[q_+])

→ Social Welfare (Q_-) = (A - D') + (B + C' + D') = (A + B) + C' & C' < C

→ Social Welfare (Q^*) = (A - D) + (B + C + D) = (A + B) + C → max!

Pecuniary benefits

- activity generates **the positive externality** (e.g. due to R&D) → the private cost curve of the activity shifts **downward**
- Q_2^* results in **higher social welfare** and higher consumer surplus (the bolded area)
- this effect is NOT external but pecuniary and results from **movement ALONG** – not a shift in – **MPB curve**



Externality optimum

Negative externality (*pollution*)

- $MPB = (MPC + \text{MEC})$
- Q^0 will lead to overproduction and $\text{Social Welfare} < \max(\text{SW})$
- Correcting **Tax** = $\text{MEC}(Q^*) \rightarrow$ reduce Q^0 to Q^*

Positive externality (*forest*)

- $(MPB + \text{MEB}) = MPC$
- Q^0 will lead to under-production and $\text{Social Welfare} < \max(\text{SW})$
- Correcting **Subsidy** = $\text{MEB}(Q^*) \rightarrow$ increase Q^0 to Q^*

where

- MPB - Marginal Private Benefits (Revenues)
- MPC – Marginal Private Cost
- MEC, MEB – Marginal External Cost, and Marginal External Benefits

How to Reach The Optimum?

The Problem

Externality as a source of market failure

- *market failure* → deviation from the first-best neoclassical world in which the price mechanism takes care of socially optimal / efficient resource allocation (Pareto *efficiency*)
 - over- (under-) production of output which generates the negative (positive) externality
 - resource allocation is not efficient > $MRS \neq MRT$ & $MRTS_i \neq MRTS_j$
- *signal failure* → market prices no longer reflect social costs (or benefits) and additional taxes (or subsidies) are called for to restore the efficient workings of the market mechanism
 - „getting prices right“
- interventions advocated by welfare economics theorems (remember Varian or see Verhoef 2002 in brief)

The Problem Solution

The unresolved tension between

- **the receptor**, who has no direct control over the size of the effect at its source, and
- **the supplier**, who has no a priori interest in the magnitude of the externality

...persist provided there is **no market on which the externality is traded**

Ronald Coase (1959)'s contribution:

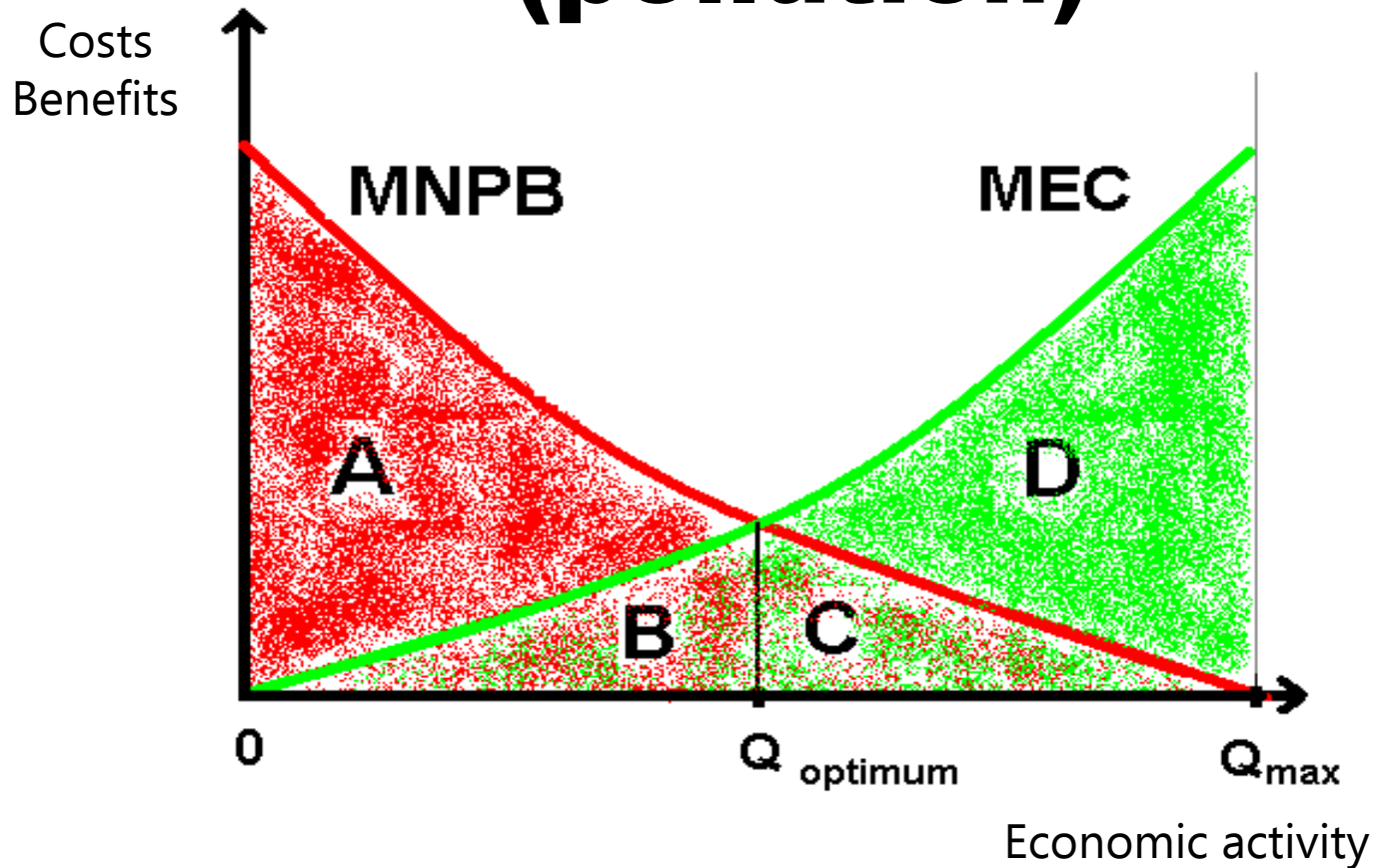
The case of radio frequencies regulation in the US:...as long as **property rights in these frequencies were well defined**, it ultimately did not matter if adjacent radio stations interfered with each other by broadcasting in the same frequency band. Furthermore, it did not matter to whom the property rights were granted. His reasoning was that **the station able to reap the higher economic gain from broadcasting would have an incentive to pay the other station not to interfere**. In the absence of transaction costs, both stations would strike a **mutually advantageous deal**.

The Problem Solution

First Coase theorem {as named by G. Stigler 1966}

- Under certain condition are fulfilled, the agents affected by the externality might reach **mutually beneficially outcome without state intervention**; if there is only one such outcome which maximizes social welfare, this outcome will be reachable by the agents **regardless how these right were assigned**
 - when **trade in an externality is possible** and there are **no transaction costs, bargaining** will lead to an efficient outcome regardless of the initial allocation of property rights
- Conditions need to be satisfied:
 - **well defined and enforced property rights**
 - **absence of transaction costs** (not prohibitively large)
 - presence of third party which enables the contract is made and **freely transferable „numeraire“ good** such as money (Stigler 1966; Acocella 1995)
 - existence of **technology** that might avoid the effect (Acocella 1995)
- transaction costs could not be neglected, and therefore, the initial allocation of property rights often mattered (Coase: *The Problem Of Social Costs*, 1960)

Optimal level of regulation (pollution)



			<i>profit</i>	<i>damage</i>	
<i>no emission</i>	→	Social Welfare (0)	=	0	− 0 = 0
<i>no regulation</i>	→	Social Welfare (Q_{max})	=	(A+B+C)	− (B+C+D) = <u>A</u> − <u>D</u>
<i>optimum</i>	→	Social Welfare (Q_{opt})	=	(A+B)	− B = A

The Problem Solution

Second Coase theorem

- If **transaction costs exist**, the possibility to reach the most efficient outcome through the market can depend on **how the property rights were initially assigned**; then, property rights should be assigned to the **actors gaining the most utility** from them.
 - However, in real life
 - nobody knows *ex ante* the most valued use of a resource
 - there exist costs involving the reallocation of resources by government
- government should create institutions which **minimize transaction costs**, so as to allow misallocations of resources to be corrected as cheaply as possible

When the Coasean solution may fail ?

transaction costs

- are usually too high for efficient bargaining to happen (J. Meade)
- would be unrealistic to assume there were no costs in the conduction of market transactions, and that these costs are "*often extremely costly, sufficiently costly at any rate to prevent many transactions that would be carried out in a world in which the pricing system worked without cost.*" (Coase, 1960)

absence of property rights”

- environmental quality is a typical “good” for which property rights are not defined and equivalently no market exists (Baumol & Oates, 1988)

prisoner’s dilemma

- the bargaining doesn't happen between two economic factors, but instead the parties might be a single large factory versus a thousand landowners nearby who might try to free ride on the other landowner’s wallets

strategic behaviour and free-riding

- In the case of undepletable externalities (Bergstrom et al. 1986; viz Verhoef 2002)

The Problem Solution

Voluntarily action via market

- Coasian bargaining
- **compensation** of an externality that does not necessarily imply its optimization because it may induce inefficient behaviour of the victims of the effect
- The externality is **internalized** through the creation of a market on which the externality is traded

Government intervention, i.e. **regulation**

- Price instruments > Pigouvian tax / subsidy (Pigou 1920) equal to MEC at the point of optimum
- Quantity-based instruments > tradable permits, or allowances
- Command-and-control measures (emission limits, technical standards)

Market-based instruments: policies and practices

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The market failure that seems most central to environmental issues is the inability of the market to address externalities from pollution

Goulder & Parry (2008)

Policy instruments: Overview

Direct regulation (DRI) (*Command-and-control measures*)

- Emission/technical standards
- Bans of toxic substances
- Land planning instruments

Market based instruments (MBI)

- Price based
 - Taxes
 - Subsidies (direct, *feed-in tariffs*,..)
 - *Bonus/malus*
- Quantity based
 - Tradable permits
 - Tradable green certificates

Cooperative instruments

- Voluntary agreements
- Roundtables
- Mediation
- Forest certification

Informational instruments

- Eco labelling
- Communication campaigns
- Education

Transferable flexible instruments

Flexibility over time

- **budgeting** of emissions or abstractions over a given period of time during which transfers may be made freely
- **banking** – transfer of permits valid for a given period to a later period(s)
- **borrowing** – advanced use of permits that are valid for a given future periods
- **differentiation** b/w normal operating regimes and short-term periods of pollution peaks

Transferable flexible instruments

Flexibility in space

- **Emission bubbles**

Designated territorial area embracing a number of sources for which a total emissions ceiling is specified for a given period of time

- **Deposition bubbles**

Applies to all sources whose emissions give rise to depositions at a given target area within a country

- **Offsets**

Transfers b/w several bubbles whose individual ceilings are adjusted accordingly (e.g. by weights)

- **Netting**

A variation of bubbles at the level of an industrial facility

Transferable flexible instruments

= tradable permits / allowances

- incentivize emissions reduction by creating a market-based trading regime - companies that produce/emit less than the amount of allowances they have been allocated can sell unused allowances to companies that have a shortage of them to reduce emissions where it is cheapest
- main types:
 - **quotas** – “**cap and trade**”, or “minimum limits and trade”
 - emission reduction credits – baseline and credit emission reduction
 - **averaging** → setting average limit values for an entire range of similar products (e.g. cars or engines)
 - **transferable rights** – transferable usage rights or rights to abstract natural resources (e.g. fishing quotas)

Tradable permits in praxis

US: SO₂ emissions

- US Clean Air Act Amendment (1990) -> Acid Rain Program to reduce SO₂ emissions predominantly from coal-burning power stations
- SO₂ trading scheme introduced starting from 1995 (Phase I)
 - number of permits to emit SO₂ gradually reduced
 - limited number of polluters, easy to monitor
 - Phase II (from 2000) – expanded to more plants
- Effectiveness:
 - SO₂ emissions reduced by >50% in 2007 (i.e. 3 years earlier than statutory deadline)
 - compliance costs ~1/4 of original predictions
 - citizen groups participation

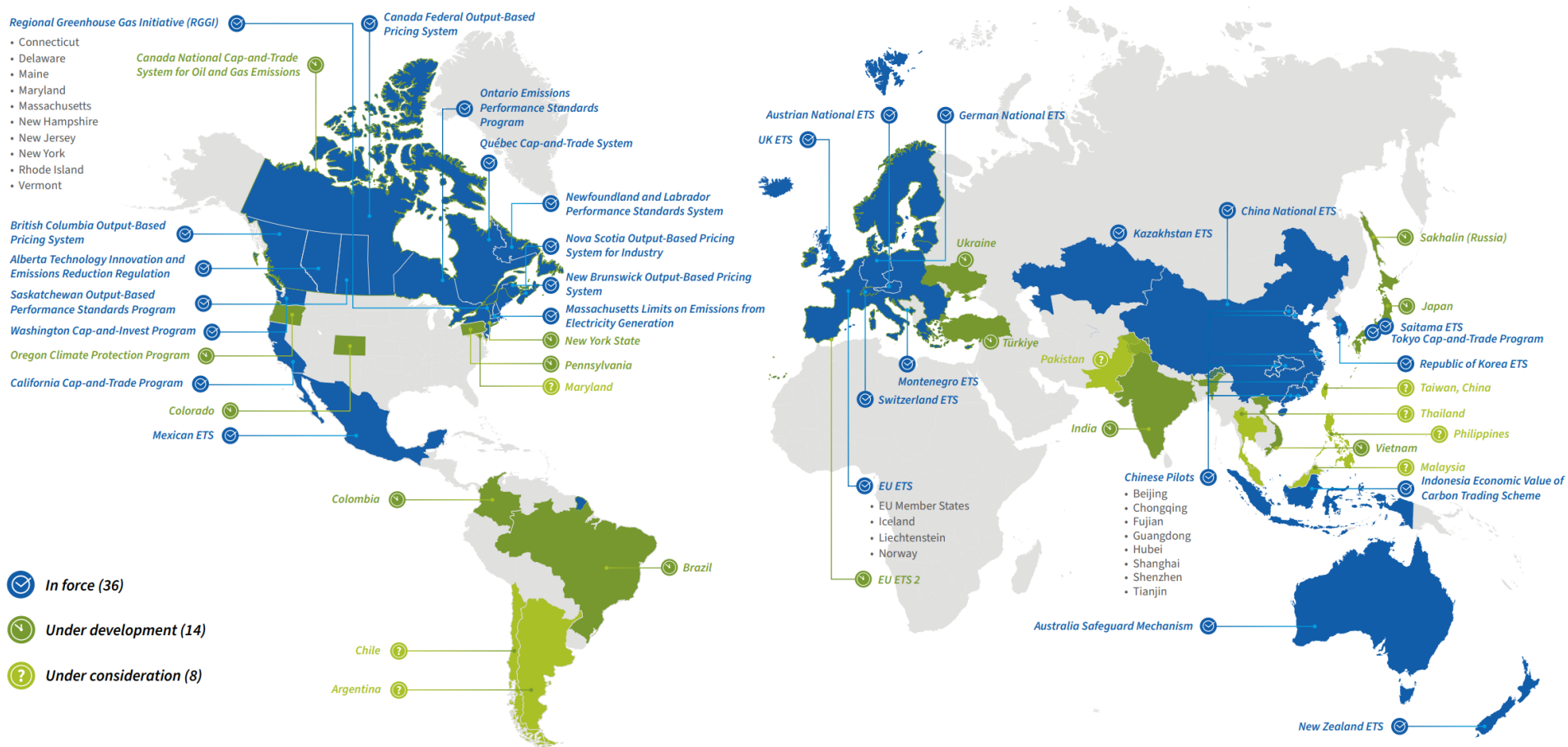
GHG (CO₂) emissions

- tradable permits now key instrument of climate change mitigation
- nowadays in EU, China, UK, some US states
- somewhat preferred to carbon taxation
 - well aligned with climate target approaches (i.e. limiting temperature increase to well below 2°C) & climate budgets (i.e. how much carbon can still be emitted)

EMISSIONS TRADING WORLDWIDE

THE CURRENT STATE OF PLAY IN CAP-AND-TRADE

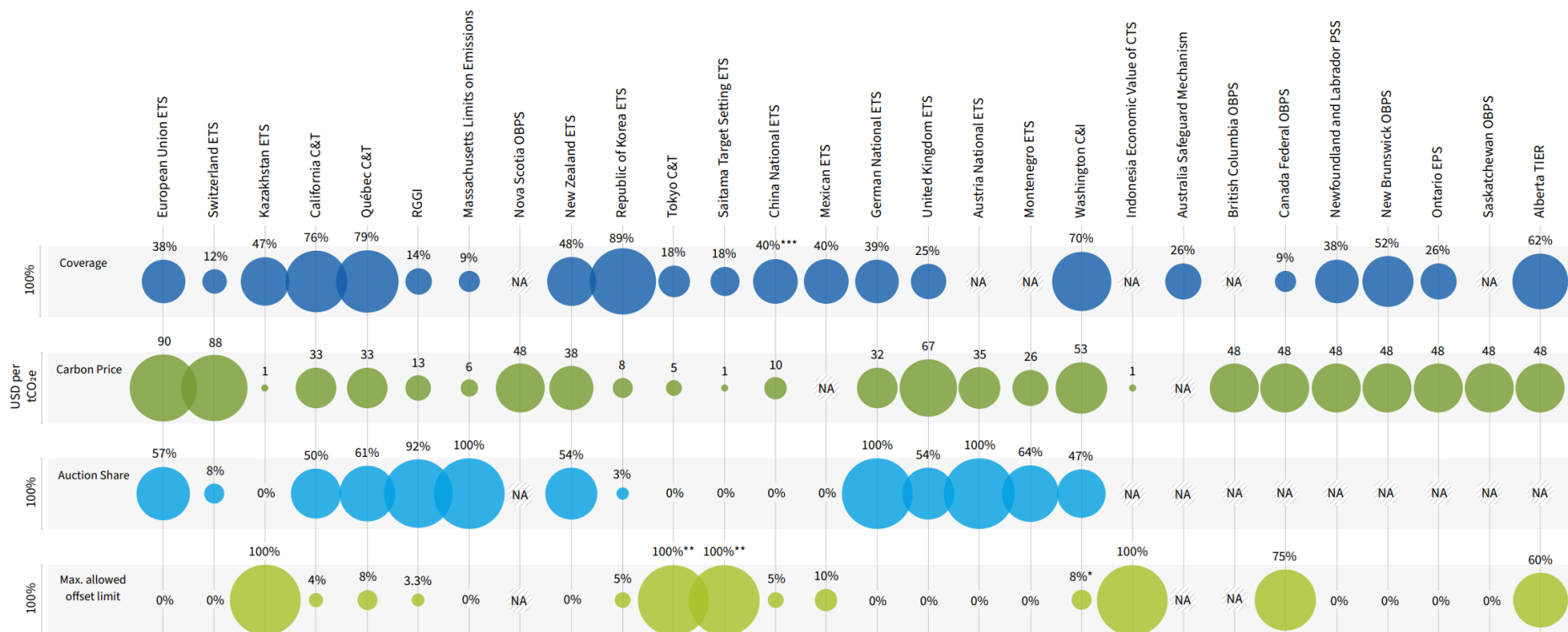
The ICAP ETS world map depicts emissions trading systems currently in force, under development or under consideration. As of January 2024, there are 36 ETSs in force. Another 14 are under development and expected to be in operation in the next few years. These include ETSs in Colombia, Türkiye, and Vietnam. 12 jurisdictions are also considering the role an ETS can play in their climate change policy mix. If a jurisdiction has multiple systems in force, it is depicted in blue, with the borders of the jurisdiction representing the layered systems (e.g. Germany and Guangdong). If, however, it has a system in force but is also developing an additional system, it is depicted in blue but also features a green border (e.g. the EU).



DIFFERENT DIMENSIONS OF ETS

A COMPARATIVE LOOK AT KEY METRICS FROM CARBON MARKETS

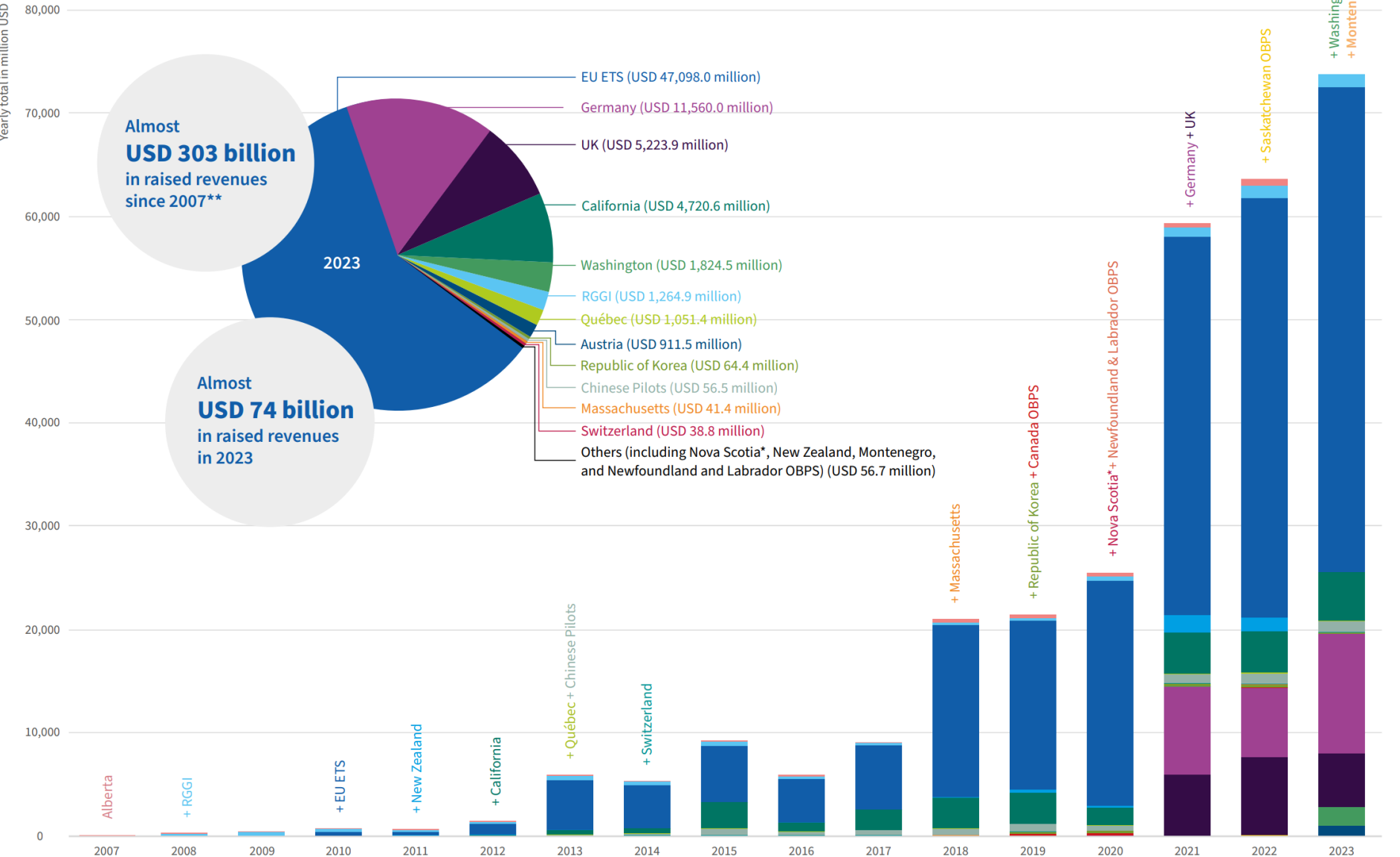
The circles below display information on different metric across ETSs in force. **Coverage** (in dark blue) shows the share of the jurisdiction's GHG emissions covered under the ETS. **Carbon price** (in dark green) is measured in USD per metric tonne of CO₂e and averaged over 2023. **Auction share** (in light blue), expressed as a share of the 2023 cap, denotes the share of allowances that have been offered for auction in the primary market. **Max. allowed offset limit** indicates the share of a compliance entity's obligations that can be met using approved offsets. The size of the circle represents the numerical value of the corresponding dimension. See "Notes on Methods and Sources" for further details.



* Up to 5% of the compliance obligation can be met with offsets. An additional 3% can be met from projects located on federally recognized tribal land.
 ** In Saitama, quantitative limits apply for "outside Saitama" credits. In Tokyo, quantitative limits apply for "outside Tokyo" credits.
 *** Of CO₂ emissions.

YEARLY REVENUES RAISED BY EACH SYSTEM

* Revenues in 2023 collected from the last auctions under the cap and trade program.
** Revenues collected by the Canada federal OBPS are reported only until 2020. Data for Newfoundland and Labrador, Saskatchewan and Alberta are only reported until 2022. Data on New Brunswick and on Ontario is not yet publicly available.



...internationally recognized

UN Framework Convention on Climate Change (UNFCCC, 1992)

- Kyoto protocol (1997, 2005)
 - GHG reduction commitments until 2020 (Kyoto II) – but just for 61 countries
 - 3 flexible mechanisms incl. international emission trading - e.g. CZ sold Kyoto permits to Japan for ~20 bln. CZK
- Paris Agreement (2015)
 - aims to **keep temperature rise well below 2°C and to limit it to 1.5°C** (relative to pre-industrial levels)
 - key measure: nationally determined contributions (**NDCs**), to be revised every 5 yrs – not formally binding („*name-and-shame*“)
 - EU was one of the first Parties to announce its post-2020 ambition
EU NDC committed to 40% emission reduction by 2030 (compared to 1990 level)
 - it is to be implemented through **EU emissions trading system**, effort-sharing for the non-ETS sectors, and regulation, support for renewable energy sources, and greater energy efficiency

EU climate-energy targets

EU 2030 climate and energy targets:

- reducing greenhouse gas emissions by 55% compared to 1990 levels,
- increasing the share of renewable energy use to 42.5%,
- improving energy efficiency by 11.7 % compared to the projected energy use for 2030 (based on the 2020 reference scenario)
 - Absolute values of primary and final energy consumption

EU 2040 climate target

Market-based instrument evaluation

Assessment of the market-based instruments:

Efficiency

Static efficiency

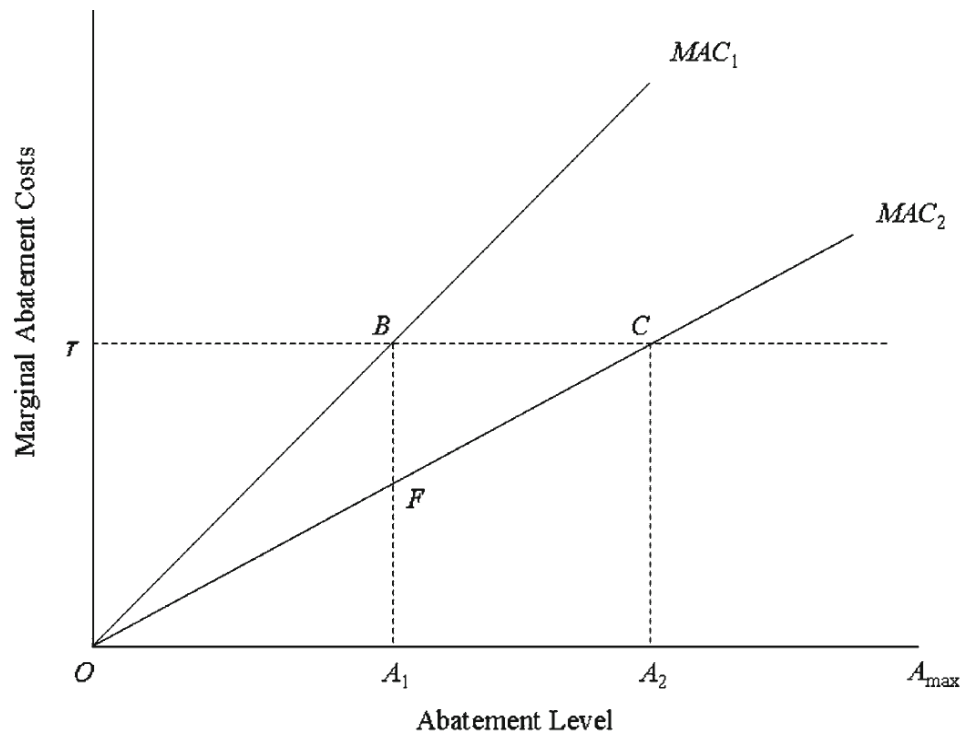
- abatement with **the least costs**
- equate **marginal abatement costs** across all potential options and agents ('static efficiency')
 - TAX: abate emissions until $MAC = TAX$
 - tradeable permit: abate and sell the permit with $P > MAC$ or emit and buy that permit for $P < MAC$
- The abatement is achieved **at minimum total costs** and **MAC are equalized** between firms

Dynamic efficiency and role of innovation

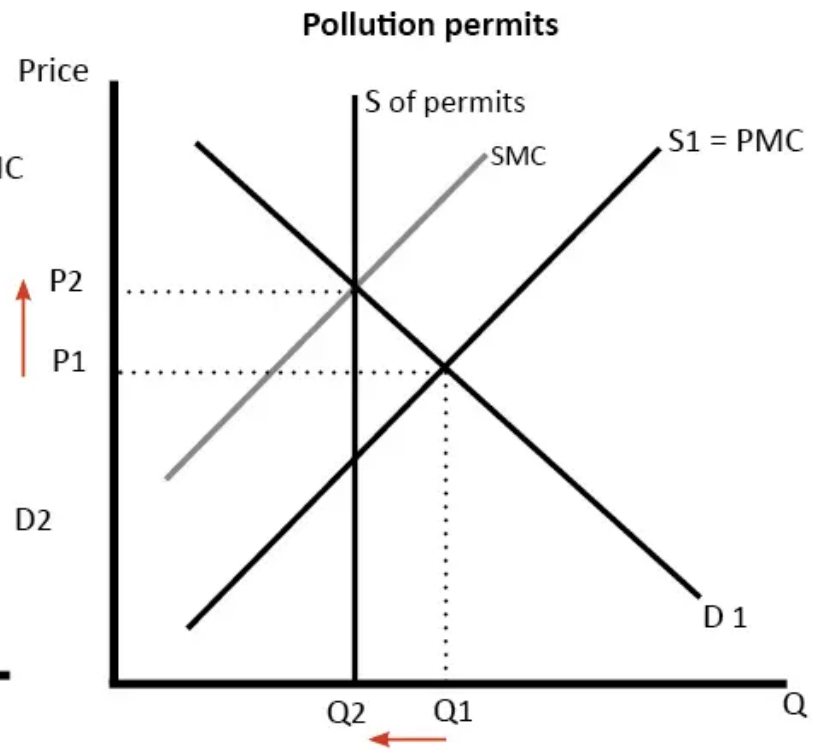
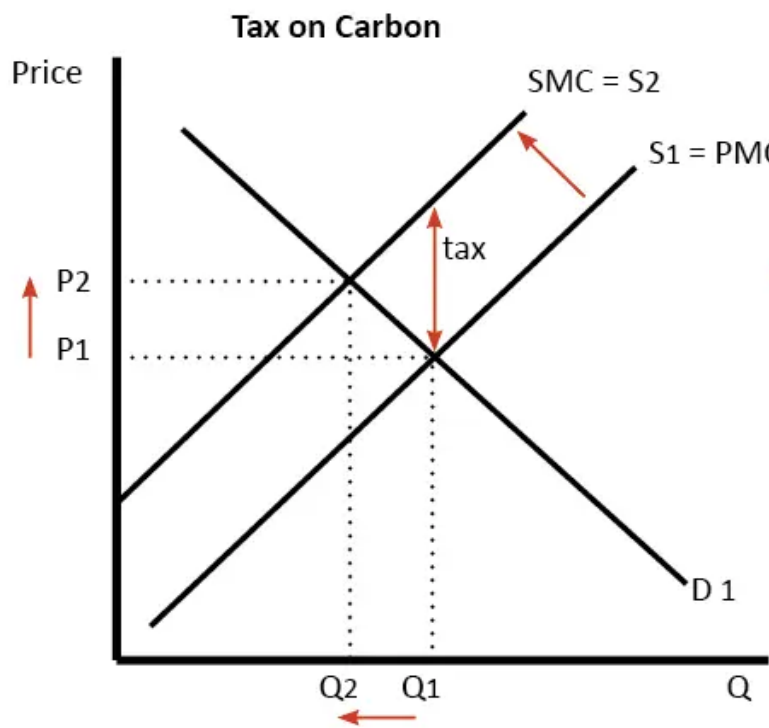
- An instrument provides **permanent incentive** ('dynamic efficiency') since each unit of emission is priced → firm likes to avoid the cost paid
- Price- vs. Quantity-based MBIs: minimize **mistake in the control**
- (Environmental) **Effectiveness**:

Dynamic efficiency – role of innovation

- Innovation in pollution control can help alleviate the tradeoff between environmental protection and industrial production (*Kneese and Schultz 1975*)
- **Marginal (dominant) approach**
 - Innovation is modeled as a reduction in MACs, e.g. $MAC_1 \rightarrow MAC_2$ in the standard graphical approach (Downing and White 1986)
 - The gains from innovation for a firm facing a Pigovian tax is asserted to be the area bounded by MAC_1 and MAC_2



- More commonly discussed only innovations **lower AC at all margins**
- Downing and White (1986) point out that innovations might reasonably **raise MACs at some margins** while lowering them at others
- Innovations need **not necessarily lead to declining MACs** (Nelissen and Requate 2004)



Assessment of the market-based instruments: (Environmental) effectiveness

- **Command-and-control | Direct measures** (a ban, 50% reduction): the target is clearly set and certain (if monitored and enforced, the target is achieved)
- **Tax | Charges | Fee** (emission, energy, carbon, resource): → the target is uncertain → depend on the tax rate (& exemptions)
- **Transferable rights** (emission allowances | permits, min quota, max quota, etc.) → the target is clear and certain
- **Other instruments** (voluntarily, information | soft measures) → difficult to enforce, and uncertain, respectively

**Is there an optimal tax /
market price of emission
allowances?**

Multi-goal of instruments (taxes)

Goal and effects of instrument

- ‘**Getting prices right**’ i.e. correct for the externalities
→ tax = Marginal External Costs at the optimum
- **Change behaviour** → motivate agents to save resources | reduce emissions
- Raise funds to finance public goods → how the revenues are recycled back to the economy

Theory on optimal taxation

- to raise a given amount of revenue while minimizing the distortions (excess burden) created by the tax system
- the first-best setting → lump-sum tax
- the second-best setting → uniform commodity taxes are rarely optimal (Ramsey 1927)
- “*inverse-elasticity rule*“, i.e. tax rates should be inversely proportional to their elasticity of demand → tax more goods with inelastic demand

Optimal environmental taxation: Sandmo (1975)

- Sandmo (1975) was the first one who divided the optimal envi tax into Ramsey part [R] and Pigou part [P], weighted by η
- Design the optimal tax system for all commodities in the commodity tax system (i.e. no taxation of labour)
 - clean commodity tax $\rightarrow$ follows the Ramsey rule
 - dirty commodity tax $\rightarrow$ composed of Ramsey component and Pigouvian component
 - η is equal to λ/μ , where λ is the marginal utility of income and μ is the effect of an increase in the tax requirement R on social utility

$$\theta_D = (1 - \eta)R + \eta P$$

Optimal environmental taxation: Sandmo (1975)

- The expression $1/\eta$ was later called ***Marginal Cost of Public Funds***
- $MCPF = -\Delta W / \Delta R$, i.e. the loss incurred by society in raising additional revenues to finance government spending

$$\theta_D = \left(1 - \frac{1}{MCPF}\right) \left(-\frac{1}{\epsilon_{DD}}\right) + \frac{1}{MCPF} \left(-N \frac{u_{ND}}{u_D}\right)$$

- If $MCPF = 1$ (i.e. individuals are indifferent between public and private consumption), the optimal tax* = the Pigou tax

Reap Double Dividend ?

Net welfare gain

- **primary welfare gain** from ecologic benefits
 - efficiency improvement in the market of the polluting goods (*Pigou*)
- **revenue-recycling effect** (efficiency gain)
 - due to the cuts in distortionary taxes
- **tax interaction effect** ('**tax interdependency effect**' – Parry 1995)
 - efficiency loss from the reduction in labour supply in response to higher pollution prices that reduces real wages (and increase demand for leisure) → are leisure and dirty goods substitutes or complements ?
 - to raise additional tax revenue on other factors of production to replace the revenue lost from the reduction in a] labour taxes, and b] revenues from enviro taxes thanks to dynamic efficiency

Double Dividend Hypothesis

First dividend ('green')

- improve the environment (see issue of effectiveness)

Second dividend ('blue')

- increase employment
- increase efficiency of the economy

Weak DD -> *generally supported*

Efficiency is improved if the revenues are recycled through cuts in distortion taxes rather than providing them lump-sum

Strong DD -> *theory not conclusive*

Overall economic efficiency is improved → the outcome is influenced by several factors such as initial structure of the tax system, degree of substitutability between and mobility of factors, wage rigidity...

Validity of weak DD

- The validity of the weak form of the DD can be theoretically proved and receives support from many numerical simulations
- However, it is subject to one assumption: there is only one distortionary tax in the economy
- Hence it holds without restrictions under the first best setting only
- The convergence of commodity tax rates is connected to the optimal taxation proposed by Ramsey (inverse elasticity rule)
- Under the weak separability assumption, optimal tax system requires equal tax rates on all commodities